Yassir CHERGUI | Data Science & AI Engineer | Master Student

Meknès, Morocco
+212 693 534 651 • cherguiyassir1@gmail.com
chergui-yassir.vercel.app • yassir-chergui • YassirCher
Availability: Immediate (Seeking 4-6 month PFE Internship)

Master's student in Data Science & AI specializing in Computer Vision, Medical AI, Generative AI, and NLP. I focus on bridging the gap between advanced research and real-world application by architecting state-of-the-art deep learning models and deploying them through scalable MLOps pipelines. Currently seeking a 4–6 month End-of-Studies Internship (PFE) in AI/ML engineering to tackle complex data challenges.

Technical Skills

- **AI & Machine Learning:** PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face,Numpy,Pandas.
- **Deep Learning Architectures:**MLP, CNN, RNN, LSTM, Transformers, Encoder-Decoder.
- **Generative AI & NLP:** LLMs (Mistral, Llama,Qwen),Embeddings, RAG, LoRA, LangChain, Prompt Engineering, LangGraph, Multi-Agent Systems, MCP .
- **Computer Vision:** YOLOv8, OpenCV, Medical Imaging (MRI), Explainable AI (Grad-CAM).
- **Databases & Big Data:** Neo4j, Cassandra, SQL, ChromaDB, Vector Databases, Semantic Search.
- **Development & MLOps:** Python, Django, Flask, FastAPI, Spring Boot, Angular, Flutter, Docker, Azure, GitHub Actions, Linux, Git.

Soft Skills

- Strong critical thinking and problem-solving mindset applied to complex ML and system design challenges.
- Results-driven and highly motivated to push the boundaries of state-of-the-art models and consistently break official benchmark scores.
- Ability to work autonomously on end-to-end AI projects, from data preparation to deployment.
- Effective collaboration and technical communication in academic and project-based teams.

Key Projects

- **Urban Security – Real-time Weapon Detection System** *YOLOv8, Django, PyTorch*
  - Developed an end-to-end real-time detection pipeline achieving >30 FPS with high performance metrics (*F1-Score: 0.9960, mAP: 0.9612*).
  - Handled data annotation, model training, and deployment via a robust REST API.
- **Fine-Tuning Mistral 7B (E-commerce Assistant)** *NLP, LLM, PEFT/LoRA, RAG*
  - Built an intelligent assistant using Parameter-Efficient Fine-Tuning (PEFT) on Mistral 7B, achieving *80.36% Mean Token Accuracy*.
  - Implemented a complete pipeline: data preprocessing, vectorization, and multi-label classification.
- **AG News Classification MLOps** *NLP, Azure, Docker, FastAPI, GitHub Actions*
  - Engineered a production-ready NLP pipeline achieving *91.34% accuracy* on the AG News dataset using Linear SVM and TF-IDF.
  - Implemented automated CI/CD via GitHub Actions for seamless Docker container deployment to Azure Container Instances (ACI).
- **Image Captioning Fine-tuning** *BLIP, PyTorch, Transformers*
  - Fine-tuned the Salesforce BLIP model on the 31k+ image Flickr30k dataset, achieving a *0.4320 ROUGE score*.
  - Optimized training efficiency on dual T4 GPUs and implemented automated evaluation using ROUGE/BLEU metrics.
- **Alzheimer MRI Classification** *TensorFlow, XAI, RAG*
  - Designed a custom CNN architecture that outperformed official benchmarks, achieving *99% classification*

accuracy.
- Integrated Grad-CAM for model explainability and RAG for medical context retrieval.

- **Medical NLP – Drug-Drug Interaction (DDI) Extraction** *BioBERT, BiLSTM, PyTorch, Flask*
  - Trained and evaluated 30 deep learning models across 5 biomedical architectures (e.g., CoTEL-D3X, BioBERT+BiLSTM), achieving a peak *93.84% accuracy* on the SemEval dataset.
  - Developed a complete Flask web application for real-time inference, model comparison, and interactive performance visualization.

- **Agentic AI Research Assistant (Multi-Agent RAG System)** *LangGraph, LangChain, FastAPI, ChromaDB, React, Docker, Azure*
  - Built a production-oriented multi-agent RAG system for document-grounded Q&A with citation support using specialized agents (Router, Retriever, Ranker, Generator, Critic).
  - Implemented asynchronous PDF ingestion, session-isolated retrieval pipelines, and vector search with ChromaDB to improve reliability and traceability.
  - Designed a full-stack containerized architecture (FastAPI + React) with Docker Compose and Nginx, enabling scalable deployment.
  - Established CI/CD pipeline with GitHub Actions for automated testing and Docker image validation, and prepared Azure deployment (ACR + Container Apps).

Certifications

- **AWS Generative AI Applications** – Coursera & AWS
- **IBM Machine Learning Professional Certificate** – IBM & Coursera
- **OCI AI Foundations Associate** – Oracle University
- **Intermediate Machine Learning** – Kaggle

Interests

AI Engineering, Generative AI, LLMs, Fine-Tuning, RAG, Transformers, NLP, Computer Vision, Medical AI, Data Science ,Machine Learning,Deep Learning.

Education

Université Moulay Ismail **Meknès**
Master of Excellence in Data Science & AI *2024–2026*
Currently in S4 (Seeking PFE). S1 & S2 Grade: Good (Bien)

Université Moulay Ismail **Meknès**
Bachelor (Licence) in SMI (Science Math Info) *2024*
Honors: Assez Bien (S5: Bien, S6: Très Bien)

High School **Meknès**
Baccalaureate in Mathematical Sciences A *2020*

Languages

English: Professional Proficiency • **French:** Professional Proficiency • **Arabic:** Native